第七章 直线章节知识点

Chapter 7 Straight Lines - Knowledge Points

本章知识点结构图 Knowledge Structure Diagram

直线方程核心公式总结 Summary of Core Formulas for Linear Equations

中文公式 Chinese Formulas	英文公式 English Formulas
倾斜角与斜率:$\alpha \in [0°, 180°)$，$k=\tanα$	Inclination Angle & Slope: $\alpha \in [0°, 180°)$, $k=\tanα$
斜率公式:$k = \frac{y_2-y_1}{x_2-x_1}(x_1 \neq x_2)$	Slope Formula: $k = \frac{y_2-y_1}{x_2-x_1}(x_1 \neq x_2)$
五种直线方程形式:1. 点斜式:$y-y_0=k(x-x_0) 2. *斜截式*: y=kx+b 3. *两点式*: = 4. *截距式*:+=15. *一般式*:Ax+By+C=0$	Five Forms of Linear Equations:1. Point-Slope: $y-y_0=k(x-x_0)$2. Slope-Intercept: $y=kx+b$3. Two-Point: $=$4. Intercept: $+=1$5. General: $Ax + By + C = 0$
平行条件:$l_1 \parallel l_2 \Leftrightarrow k_1 = k_2$(斜率存在)	Parallelism Condition: $l_1 \parallel l_2 \Leftrightarrow k_1 = k_2$ (slopes exist)
垂直条件:$l_1 \perp l_2 \Leftrightarrow k_1 \cdot k_2 = -1$(斜率存在)	Perpendicularity Condition: $l_1 \perp l_2 \Leftrightarrow k_1 \cdot k_2 = -1$ (slopes exist)
交点:方程组$\begin{cases} A_1x + B_1y + C_1 = 0 \\ A_2x + B_2y + C_2 = 0 \end{cases}$的解	Intersection: Solution of the system $\begin{cases} A_1x + B_1y + C_1 = 0 \\ A_2x + B_2y + C_2 = 0 \end{cases}$

7.1 直线的倾斜角和斜率 7.1 Inclination Angle and Slope of a Straight Line

一、基础知识 I. Basic Knowledge

1. 核心定义 1. Core Definitions

- **倾斜角(Inclination Angle)**:

  中文:平面直角坐标系中，x 轴正方向与直线 l 向上方向之间的最小正角，记为 α，范围是$0° \leq \alpha < 180°$(或$0 \leq \alpha < \pi$弧度)。

  English: In the rectangular coordinate system, the smallest positive angle between the positive direction of the x-axis and the upward direction of the straight line l, denoted as α, with the range $0° \leq \alpha < 180°$ (or $0 \leq \alpha < \pi$ radians).

- **斜率(Slope)**:

  中文:倾斜角 α 的正切值，记为 k，即$k=\tanα$。

  English: The tangent value of the inclination angle α, denoted as k, i.e., $k=\tanα$.

2. 斜率公式推导 2. Derivation of Slope Formula

中文:设直线 l 经过两点$P_1(x_1,y_1)$和$P_2(x_2,y_2)(x_1 \neq x_2)$，过$P_1$作 x 轴平行线，过$P_2$作 y 轴平行线，交于点$Q(x_2,y_1)$，则:

- $P_1Q = |x_2 - x_1|$(水平距离)
- $QP_2 = |y_2 - y_1|$(垂直距离)
- 由三角函数定义:$\tanα=\frac{QP_2}{P_1Q}=\frac{y_2-y_1}{x_2-x_1}$(符号由 α 的锐角 / 钝角决定，与差值符号一致)

  最终得斜率公式:$k = \frac{y_2-y_1}{x_2-x_1}(x_1 \neq x_2)$

English: Let line l pass through two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)(x_1 \neq x_2)$. Draw a line parallel to the x-axis through P_1 and a line parallel to the y-axis through P_2, intersecting at point $Q(x_2, y_1)$. Then:

- $P_1Q = |x_2 - x_1|$ (horizontal distance)
- $QP_2 = |y_2 - y_1|$ (vertical distance)
- By trigonometric definition: $\tanα=\frac{QP_2}{P_1Q}=\frac{y_2-y_1}{x_2-x_1}$ (the sign is determined by whether α is acute/obtuse, consistent with the sign of the difference)

  Finally, the slope formula is obtained: $k = \frac{y_2-y_1}{x_2-x_1}(x_1 \neq x_2)$

3. 特殊情况说明 3. Special Cases

倾斜角 α (Inclination Angle)	直线特征 (Line Feature)	斜率 k 的取值 (Value of Slope k)
$0°$	平行于 x 轴(水平直线)(Parallel to x-axis, horizontal line)	$k = 0$
$0° < \alpha < 90°$	锐角倾斜 (Acute inclination)	$k > 0$(α 越大，k 越大)(The larger α is, the larger k is)
$90°$	垂直于 x 轴(竖直直线)(Perpendicular to x-axis, vertical line)	斜率不存在 (No slope)
$90° < \alpha < 180°$	钝角倾斜 (Obtuse inclination)	$k < 0$(α 越大，k 越接近 0)(The larger α is, the closer k is to 0)

二、例题 II. Examples

例题 1

中文:求经过点$A(2,5)$和$B(4,1)$的直线的斜率和倾斜角。

English: Find the slope and inclination angle of the straight line passing through points $A(2,5)$ and $B(4,1)$.

解(Solution):

1. 代入斜率公式 (Substitute into the slope formula):

   $k = \frac{y_2-y_1}{x_2-x_1} = \frac{1-5}{4-2} = \frac{-4}{2} = -2$

2. 求倾斜角 α (Find the inclination angle α):

   由$k=\tanα=-2$，且$90° < \alpha < 180°$(因 k<0)(Since $k=\tanα=-2$ and $90° < \alpha < 180°$ (because k<0)),

   得$\alpha = 180° - \arctan2 \approx 180° - 63.43° = 116.57°$(或$\pi - \arctan2$弧度)(Thus, $\alpha = 180° - \arctan2 \approx 180° - 63.43° = 116.57°$ (or $\pi - \arctan2$ radians))

答案(Answer):斜率为-2，倾斜角约为$116.57°$(Slope = -2, Inclination Angle ≈ $116.57°$)

⚠ 易错点解析⚠ Common Mistakes

1. 忽略倾斜角范围:误将 α 取为$\arctan(-k)$(负角)，需修正为$180° - \arctan|k|$;

   Ignoring the range of inclination angle: Mistakenly taking α as $\arctan(-k)$ (negative angle), which should be corrected to $180° - \arctan|k|$;

2. 斜率公式误用:当$x_1 = x_2$时，强行代入公式导致分母为 0，需直接判断斜率不存在;

   Misusing the slope formula: When $x_1 = x_2$, forcing substitution into the formula results in a denominator of 0; directly judge that the slope does not exist;

3. 混淆“水平直线”与“竖直直线”:$y=3$是水平直线(α=0°，k=0)，$x=-2$是竖直直线(α=90°，无斜率)。

Confusing “horizontal line” with “vertical line”: $y=3$ is a horizontal line (α=0°, k=0), and $x=-2$ is a vertical line (α=90°, no slope).

拓展例题 Extended Example

例题 2

中文:已知直线 l 的倾斜角是直线$y=\frac{1}{\sqrt{3}}x+2$倾斜角的 2 倍，求直线 l 的斜率。

English: Given that the inclination angle of line l is twice that of the line $y=\frac{1}{\sqrt{3}}x+2$, find the slope of line l.

解(Solution):

设直线$y=\frac{1}{\sqrt{3}}x+2$的倾斜角为 α，则$\tanα=\frac{1}{\sqrt{3}}$，故$\alpha=30°$。

Let the inclination angle of the line $y=\frac{1}{\sqrt{3}}x+2$ be α, then $\tanα=\frac{1}{\sqrt{3}}$, so $\alpha=30°$.

直线 l 的倾斜角为$2\alpha=60°$，则斜率$k=\tan 60°=\sqrt{3}$。

The inclination angle of line l is $2\alpha=60°$, so the slope $k=\tan 60°=\sqrt{3}$.

答案(Answer):$k=\sqrt{3}$

7.2 直线的方程 7.2 Forms of Linear Equations

一、基础知识 I. Basic Knowledge

1. 直线方程的五种形式 1. Five Forms of Linear Equations

形式名称 (Form Name)	已知条件 (Given Conditions)	方程表达式 (Equation)	适用范围 (Scope of Application)	优缺点 (Advantages & Disadvantages)
点斜式 (Point-Slope Form)	过点$P(x_0,y_0)$，斜率 k (Passes through $P(x_0,y_0)$ with slope k)	$y-y_0=k(x-x_0)$	斜率存在 (α≠90°) (Slope exists)	优点:直接代入点和斜率；缺点:斜率不存在时不适用 (Advantage: Directly substitute point and slope; Disadvantage: Not applicable when slope does not exist)
斜截式 (Slope-Intercept Form)	斜率 k, y 轴截距 b(与 y 轴交点 (0,b)) (Slope k, y-intercept b (intersection with y-axis at (0,b)))	$y=kx+b$	斜率存在 (α≠90°) (Slope exists)	优点:直观体现斜率和 y 截距；缺点:斜率不存在时不适用 (Advantage: Intuitively reflects slope and y-intercept; Disadvantage: Not applicable when slope does not exist)
两点式 (Two-Point Form)	过两点$A(x_1,y_1)$、$B(x_2,y_2)$ (Passes through two points $A(x_1,y_1)$ and $B(x_2,y_2)$)	$\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}$	$x_1\neq x_2$且$y_1\neq y_2$	优点:直接用两点坐标；缺点:垂直 x 轴 /y 轴时不适用 (Advantage: Directly uses coordinates of two points; Disadvantage: Not applicable when perpendicular to x-axis/y-axis)
截距式 (Intercept Form)	x 轴截距 a(与 x 轴交点 (a,0))，y 轴截距 b(与 y 轴交点 (0,b)) (x-intercept a (intersection with x-axis at (a,0)), y-intercept b (intersection with y-axis at (0,b)))	$\frac{x}{a}+\frac{y}{b}=1$	$a\neq 0$且$b\neq 0$(不过原点) (Does not pass through the origin)	优点:便于画图；缺点:过原点或截距为 0 时不适用 (Advantage: Easy to plot; Disadvantage: Not applicable when passing through the origin or intercept is 0)
一般式 (General Form)	无特殊条件(A、B 不同时为 0) (No special conditions (A and B are not zero simultaneously))	$Ax+By+C=0$ $(A^2+B^2\neq 0)$	所有直线通用 (Applicable to all lines)	优点:通用性强；缺点:不直观体现斜率和截距 (Advantage: High versatility; Disadvantage: Does not intuitively reflect slope and intercept)

2. 公式推导 2. Formula Derivation

- **点斜式推导**:由斜率公式$k=\frac{y-y_0}{x-x_0}(x\neq x_0)$，两边同乘$(x-x_0)$得$y-y_0=k(x-x_0)$；

  Derivation of Point-Slope Form: From the slope formula $k=\frac{y-y_0}{x-x_0}(x\neq x_0)$, multiply both sides by $(x-x_0)$ to get $y-y_0=k(x-x_0)$;

- **斜截式推导**:点斜式中取点$(0,b)$(y 轴截距)，代入得$y-b=k(x-0)$，即$y=kx+b$；

  Derivation of Slope-Intercept Form: Take the point $(0,b)$ (y-intercept) in the point-slope form, substitute to get $y-b=k(x-0)$, i.e., $y=kx+b$;

- **一般式转换**:任意形式均可整理为$Ax+By+C=0$，例如斜截式$y=kx+b$可化为$kx-y+b=0$(此时$A=k$，$B=-1$，$C=b$)。

  Conversion to General Form: Any form can be rearranged into $Ax+By+C=0$. For example, the slope-intercept form $y=kx+b$ can be converted to $kx-y+b=0$ (where $A=k$, $B=-1$, $C=b$).

二、例题 II. Examples

例题 1(点斜式)

中文:求经过点$P(3,-2)$且斜率为 3 的直线方程，并化为一般式。

English: Find the equation of the line passing through $P(3,-2)$ with slope 3, and convert it to the general form.

解(Solution):

代入点斜式$y - y_0 = k(x - x_0)$，得 (Substitute into the point-slope form $y - y_0 = k(x - x_0)$):

$$y - (-2) = 3(x - 3) \Rightarrow y + 2 = 3x - 9$$

整理为一般式 (Rearrange into general form): $3x - y - 11 = 0$

答案(Answer):$3x - y - 11 = 0$

例题 2(斜截式)

中文:求斜率为-2且 y 轴截距为 4 的直线方程，化为截距式。

English: Find the equation of the line with slope -2 and y-intercept 4, and convert it to the intercept form.

解(Solution):

代入斜截式$y = kx + b$，得 (Substitute into the slope-intercept form $y = kx + b$):

$$y = -2x + 4$$

整理为截距式 (Rearrange into intercept form): $\frac{x}{2} + \frac{y}{4} = 1$(令 x=0 得 y=4，令 y=0 得 x=2，即 a=2，b=4)(Let x=0 to get y=4, let y=0 to get x=2, i.e., a=2, b=4)

答案(Answer):$\frac{x}{2} + \frac{y}{4} = 1$

7.3 直线的平行和垂直 7.3 Parallelism and Perpendicularity of Straight Lines

一、基础知识 I. Basic Knowledge

1. 核心条件 1. Core Conditions

设两条不重合的直线l_1和l_2，斜率分别为k_1和k_2(若斜率存在)，倾斜角分别为α_1和α_2。

Let there be two non-coincident lines l_1 and l_2, with slopes k_1 and k_2 (if slopes exist) and inclination angles α_1 and α_2 respectively.

(1) 平行(Parallelism, $l_1 \parallel l_2$)

- **推导(Derivation)**:

  中文:平行直线倾斜角相等($\alpha_1 = \alpha_2$)，故$\tan\alpha_1 = \tan\alpha_2$，即$k_1 = k_2$；若斜率不存在($\alpha_1 = \alpha_2 = 90°$)，两直线均垂直 x 轴，也平行。

  English: Parallel lines have equal inclination angles ($\alpha_1 = \alpha_2$), so $\tan\alpha_1 = \tan\alpha_2$, i.e., $k_1 = k_2$; if slopes do not exist ($\alpha_1 = \alpha_2 = 90°$), both lines are perpendicular to the x-axis and are also parallel.

- **条件总结(Condition Summary)**:

$$\boxed{l_1 \parallel l_2 \Leftrightarrow \begin{cases} k_1 = k_2 \quad \text{(斜率存在)} \\ \text{两直线均垂直x轴} \quad \text{(斜率不存在)} \end{cases}}$$

(2) 垂直(Perpendicularity, $l_1 \perp l_2$)

- **推导(Derivation)**:

  中文:垂直直线倾斜角相差90°($\alpha_2=\alpha_1+90^\circ$),故$\tan\alpha_2=\tan(\alpha_1+90^\circ)=-\cot\alpha_1=-\frac{1}{\tan\alpha_1}$,即$k_1\cdot k_2=-1$;若一条斜率为 0($\alpha_1=0^\circ$,平行 x 轴),另一条斜率不存在($\alpha_2=90^\circ$,垂直 x 轴),两直线也垂直。

  English: Perpendicular lines have inclination angles differing by 90° ($\alpha_2=\alpha_1+90^\circ$), so $\tan\alpha_2=\tan(\alpha_1+90^\circ)=-\cot\alpha_1=-\frac{1}{\tan\alpha_1}$, i.e., $k_1\cdot k_2=-1$; if one line has a slope of 0 ($\alpha_1=0^\circ$, parallel to the x-axis) and the other has no slope ($\alpha_2=90^\circ$, perpendicular to the x-axis), the two lines are also perpendicular.

- **条件总结(Condition Summary)**:

$$\boxed{l_1\perp l_2 \Leftrightarrow \begin{cases} k_1\cdot k_2=-1 \quad (\text{斜率存在}) \\ \text{一条斜率为0,另一条斜率不存在} \end{cases}}$$

(3) 重合(Coincidence)

- **条件(Condition)**:

  中文:$k_1=k_2$且两直线过同一点(或斜截式中$b_1=b_2$)。

  English: $k_1=k_2$ and the two lines pass through the same point (or $b_1=b_2$ in slope-intercept form).

2. 一般式条件 2. Conditions in General Form

设$l_1: A_1x+B_1y+C_1=0$,$l_2: A_2x+B_2y+C_2=0$($A_1^2+B_1^2\neq 0$,$A_2^2+B_2^2\neq 0$):

Let $l_1: A_1x+B_1y+C_1=0$ and $l_2: A_2x+B_2y+C_2=0$ $(A_1^2+B_1^2\neq 0,\ A_2^2+B_2^2\neq 0)$:

位置关系(Positional Relationship)	一般式条件(Condition in General Form)
平行(Parallel，$l_1 \parallel l_2$)	$A_1B_2=A_2B_1$ 且 $A_1C_2\neq A_2C_1$(避免重合，to avoid coincidence)
垂直(Perpendicular，$l_1 \perp l_2$)	$A_1A_2+B_1B_2=0$
重合(Coincident)	$A_1B_2=A_2B_1$ 且 $A_1C_2=A_2C_1$
相交(Intersecting)	$A_1B_2\neq A_2B_1$

二、例题 II. Examples

例题 1(垂直条件)

中文:直线$l_1: 3x-ay=2$和直线$l_2: x+2y=5$垂直，求实数 a 的值。

English: Line $l_1: 3x-ay=2$ is perpendicular to line $l_2: x+2y=5$, find the value of real number a.

解(Solution):

将两直线化为斜截式求斜率(Convert both lines to slope-intercept form to find slopes):

- $l_1: y=\frac{3}{a}x-\frac{2}{a}$，故$k_1=\frac{3}{a}(a\neq 0)$；$l_1: y=\frac{3}{a}x-\frac{2}{a}$, so $k_1=\frac{3}{a}(a\neq 0)$;
- $l_2: y=-\frac{1}{2}x+\frac{5}{2}$，故$k_2=-\frac{1}{2}$。$l_2: y=-\frac{1}{2}x+\frac{5}{2}$, so $k_2=-\frac{1}{2}$.

由垂直条件$k_1\cdot k_2=-1$(From the perpendicularity condition $k_1\cdot k_2=-1$):

$$\frac{3}{a}\times\left(-\frac{1}{2}\right)=-1\Rightarrow-\frac{3}{2a}=-1\Rightarrow 2a=3\Rightarrow a=\frac{3}{2}$$

答案(Answer):$a=\frac{3}{2}$

例题 2(平行条件)

中文:直线l_1经过点$(5,3)$且与直线$l_2: 4x-6y+1=0$平行，求直线l_1的方程。

English: Line l_1 passes through $(5,3)$ and is parallel to line $l_2: 4x-6y+1=0$, find the equation of line l_1.

解(Solution):

1. 求l_2的斜率(Find the slope of l_2):$l_2: 4x-6y+1=0\Rightarrow y=\frac{2}{3}x+\frac{1}{6}$, 故$k_2=\frac{2}{3}$。$l_2: 4x-6y+1=0\Rightarrow y=\frac{2}{3}x+\frac{1}{6}$, so $k_2=\frac{2}{3}$.
2. 由平行条件，l_1的斜率$k_1=k_2=\frac{2}{3}$(From the parallelism condition, the slope of l_1 is $k_1=k_2=\frac{2}{3}$)。
3. 代入点斜式(过点$(5,3)$)(Substitute into the point-slope form (passes through $(5,3)$)):

$$y-3=\frac{2}{3}(x-5)$$

整理为一般式(Rearrange into general form):$2x-3y-1=0$

答案(Answer):$2x-3y-1=0$

⚠ 易错点解析 ⚠ Common Mistakes

1. 平行条件遗漏"不重合":仅$k_1 = k_2$可能是重合直线，需验证截距不同(斜截式)或$A_1C_2 \neq A_2C_1$(一般式);

   Omitting "non-coincident" in the parallelism condition: $k_1 = k_2$ alone may indicate coincident lines; verify that intercepts are different (slope-intercept form) or $A_1C_2 \neq A_2C_1$ (general form);

2. 垂直条件忽略斜率不存在:例如直线$x = 1$(无斜率)与$y = 2$(斜率 0)是垂直的，不能用$k_1 \cdot k_2 = -1$判断;

   Ignoring "no slope" in the perpendicularity condition: For example, line $x = 1$ (no slope) and line $y = 2$ (slope = 0) are perpendicular, which cannot be judged by $k_1 \cdot k_2 = -1$;

3. 一般式转换错误:求斜率时误将$Ax + By + C = 0$化为$y = -\frac{B}{A}x - \frac{C}{A}$(正确应为$y = -\frac{A}{B}x - \frac{C}{B}$, $B \neq 0$);

   Mistake in converting to general form: When finding the slope, incorrectly converting $Ax + By + C = 0$ to $y = -\frac{B}{A}x - \frac{C}{A}$ (the correct form is $y = -\frac{A}{B}x - \frac{C}{B}$, $B \neq 0$);

拓展例题 Extended Example

中文:已知直线$l: (m^2 - 1)x + (m + 1)y - 4 = 0$, 分别求 m 的值使得:(1) l 与 x 轴平行; (2) l 与直线$2x - y + 1 = 0$垂直。

English: Given line $l: (m^2 - 1)x + (m + 1)y - 4 = 0$, find the value of m such that: (1) l is parallel to the x-axis; (2) l is perpendicular to the line $2x - y + 1 = 0$.

解(Solution):

(1) 与 x 轴平行的直线斜率为 0，故(A line parallel to the x-axis has a slope of 0, so):

$$\begin{cases} m^2 - 1 = 0 & (\text{x 系数为 0，保证斜率为 0}) \\ m + 1 \neq 0 & (\text{y 系数不为 0，避免直线不存在}) \end{cases}$$

解得$m = 1$($m = -1$时 y 系数为 0，直线方程为$-4 = 0$，无解)。

Solve to get $m = 1$ (when $m = -1$, the y-coefficient is 0, and the line equation becomes $-4 = 0$, which has no solution).

(2) 直线$2x - y + 1 = 0$的斜率为 2，由垂直条件(The slope of line $2x - y + 1 = 0$ is 2; from the perpendicularity condition):$k_l \cdot 2 = -1 \Rightarrow k_l = -\frac{1}{2}$

直线 l 的斜率$k_l = -\frac{m^2-1}{m+1} = -\frac{1}{2}(m + 1 \neq 0)$(The slope of line l is $k_l = -\frac{m^2-1}{m+1} = -\frac{1}{2}(m + 1 \neq 0)$),

化简(Simplify):

$$\frac{(m-1)(m+1)}{m+1} = \frac{1}{2} \Rightarrow m - 1 = \frac{1}{2} \Rightarrow m = \frac{3}{2}$$

答案(Answer):(1) $m = 1$；(2) $m = \frac{3}{2}$

7.4 两条直线的交点 7.4 Intersection of Two Straight Lines

一、基础知识 I. Basic Knowledge

1. 交点定义 1. Definition of Intersection

中文:两条直线的交点坐标，是它们的方程组成的方程组的解(坐标满足两条直线的方程)。

English: The intersection coordinate of two lines is the solution to the system of equations formed by their equations (the coordinate satisfies both line equations).

若方程组$\begin{cases}A_1x+B_1y+C_1=0\\A_2x+B_2y+C_2=0\end{cases}$:

For the system of equations $\begin{cases}A_1x+B_1y+C_1=0\\A_2x+B_2y+C_2=0\end{cases}$:

- 有唯一解 → 两直线相交(intersect)，解为交点坐标;

  Has a unique solution → the two lines intersect, and the solution is the intersection coordinate;

- 无解 → 两直线平行(parallel);

  Has no solution → the two lines are parallel;

- 有无穷多解 → 两直线重合(coincident)。

  Has infinitely many solutions → the two lines are coincident.

2. 位置关系总结表 2. Summary Table of Positional Relationships

位置关系 (Positional Relationship)	斜率形式(Slope Form) $l_1: y = k_1x + b_1$； $l_2: y = k_2x + b_2$	一般式形式(General Form)$l_1: A_1x + B_1y + C_1 = 0$; $l_2: A_2x + B_2y + C_2 = 0$
相交 (Intersecting)	$k_1 \neq k_2$	$A_1B_2 \neq A_2B_1$ (方程组有唯一解， system has a unique solution)
平行 (Parallel)	$k_1 = k_2$ 且 $b_1 \neq b_2$	$A_1B_2 = A_2B_1$ 且 $A_1C_2 \neq A_2C_1$ (方程组无解，system has no solution)
重合 (Coincident)	$k_1 = k_2$ 且 $b_1 = b_2$	$A_1B_2 = A_2B_1$ 且 $A_1C_2 = A_2C_1$(方程组无穷多解，system has infinitely many solutions)
垂直 (Perpendicular)	$k_1 \cdot k_2 = -1$(或一斜率为 0，一斜率不存在, or one slope = 0, the other has no slope)	$A_1A_2 + B_1B_2 = 0$

二、例题 II. Examples

例题 1(求交点坐标)

中文:求直线$2x + 5y - 7 = 0$和$3x - 2y + 1 = 0$的交点坐标。

English: Find the intersection coordinate of the lines $2x+5y-7=0$ and $3x-2y+1=0$.

解(Solution):

解方程组(Solve the system of equations)$\begin{cases}2x+5y=7 & (1)\\ 3x-2y=-1 & (2)\end{cases}$

用消元法(Using elimination method):

(1)×3 - (2)×2 → $6x+15y-6x+4y=21+2 \Rightarrow 19y=23 \Rightarrow y=\frac{23}{19}$

代入 (2)(Substitute into (2)):

$$3x-2\times\frac{23}{19}=-1 \Rightarrow 3x=\frac{46}{19}-1=\frac{27}{19} \Rightarrow x=\frac{9}{19}$$

答案(Answer):交点坐标为$\left(\frac{9}{19},\frac{23}{19}\right)$(Intersection coordinate is $\left(\frac{9}{19},\frac{23}{19}\right)$)

例题 2(综合应用)

中文:求经过直线$2x+5y-7=0$和$3x-2y+1=0$的交点，且垂直于直线$l_1: 5x-y+3=0$的直线l_2的方程。

English: Find the equation of line l_2 that passes through the intersection of $2x+5y-7=0$ and $3x-2y+1=0$, and is perpendicular to line $l_1: 5x-y+3=0$.

解(Solution):

1. 由例题 1 得交点$P\left(\frac{9}{19},\frac{23}{19}\right)$(From Example 1, the intersection point is $P\left(\frac{9}{19},\frac{23}{19}\right)$);

2. 求l_1的斜率(Find the slope of l_1):$l_1: 5x - y + 3 = 0 \Longrightarrow y = 5x + 3$，故$k_1 = 5$；$l_1: 5x - y + 3 = 0 \Longrightarrow y = 5x + 3$, so $k_1 = 5$;
3. 由垂直条件，l_2的斜率$k_2 = -\frac{1}{k_1} = -\frac{1}{5}$(From the perpendicularity condition, the slope of l_2 is $k_2 = -\frac{1}{k_1} = -\frac{1}{5}$);
4. 代入点斜式(过点 P)(Substitute into the point-slope form (passes through point P)):

$$y - \frac{23}{19} = -\frac{1}{5}\left(x - \frac{9}{19}\right)$$

两边同乘 95(19 和 5 的最小公倍数)消分母(Multiply both sides by 95 (LCM of 19 and 5) to eliminate denominators):

$$95y - 115 = -19x + 9 \Longrightarrow 19x + 95y - 124 = 0$$

答案(Answer):直线l_2的方程为$19x + 95y - 124 = 0$(The equation of line l_2 is $19x + 95y - 124 = 0$)

⚠ 易错点解析⚠ Common Mistakes

1. 解方程组出错:消元时符号错误，或代入计算失误，建议用两种方法验证(代入法 + 消元法);

   Error in solving the system of equations: Sign errors during elimination or miscalculations during substitution; it is recommended to verify with two methods (substitution method + elimination method);

2. 混淆 “过交点” 与 “过定点”:过两直线交点的直线可设为$A_1x + B_1y + C_1 + \lambda(A_2x + B_2y + C_2) = 0$($\lambda$为参数)，避免求交点的复杂计算;

   Confusing “passing through the intersection” with “passing through a fixed point”: A line passing through the intersection of two lines can be expressed as $A_1x+B_1y+C_1+\lambda(A_2x+B_2y+C_2)=0(\lambda$ is a parameter), avoiding complex calculations of finding the intersection point;

3. 忽略直线存在性:例如设过交点的直线方程时，需确保系数不同时为 0;

   Ignoring the existence of the line: For example, when setting the equation of a line passing through the intersection, ensure that the coefficients are not zero simultaneously;

拓展例题 Extended Example

中文:求过直线$x+y-2=0$和$x-2y+1=0$的交点，且到点$P(2,1)$的距离为$\sqrt{2}$的直线方程。

English: Find the equation of the line that passes through the intersection of $x+y-2=0$ and $x-2y+1=0$, and has a distance of $\sqrt{2}$ to point $P(2,1)$.

解(Solution):

1. 设过交点的直线方程为$x+y-2+\lambda(x-2y+1)=0(\lambda$为参数)，整理为(Let the equation of the line passing through the intersection be $x+y-2+\lambda(x-2y+1)=0(\lambda$ is a parameter), and rearrange it to):

$$(1+\lambda)x+(1-2\lambda)y+(\lambda-2)=0$$

2. 由点到直线的距离公式(From the distance formula from a point to a line):

$$\frac{|(1+\lambda)\times 2+(1-2\lambda)\times 1+(\lambda-2)|}{\sqrt{(1+\lambda)^2+(1-2\lambda)^2}}=\sqrt{2}$$

3. 化简分子(Simplify the numerator):$|2+2\lambda+1-2\lambda+\lambda-2|=|\lambda+1|$

   分母(Denominator):$\sqrt{1+2\lambda+\lambda^2+1-4\lambda+4\lambda^2}=\sqrt{5\lambda^2-2\lambda+2}$

4. 列方程(Set up the equation):

$$\frac{|\lambda+1|}{\sqrt{5\lambda^2-2\lambda+2}}=\sqrt{2}$$

两边平方(Square both sides):

$$(\lambda+1)^2=2(5\lambda^2-2\lambda+2)\Longrightarrow\lambda^2+2\lambda+1=10\lambda^2-4\lambda+4\Longrightarrow9\lambda^2-6\lambda+3=0$$

解得$\lambda=1$或$\lambda=\frac{1}{3}$(Solve to get $\lambda=1$ or $\lambda=\frac{1}{3}$)

5. 代入直线方程(Substitute into the line equation):

- 当$\lambda=1$时(When $\lambda=1$):$2x-y-1=0$;
- 当$\lambda=\frac{1}{3}$时(When $\lambda=\frac{1}{3}$):$4x+y-5=0$

答案(Answer):直线方程为$2x-y-1=0$或$4x+y-5=0$

(The line equations are $2x-y-1=0$ or $4x+y-5=0$)
